GITAM Deemed to be University

Mechanical Engineering | Course Code: ME304

Heat & Mass Transfer - Laboratory Manual & Practical Guide

Syllabus Focus: ME304 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Steady & Transient Conduction

" Syllabus Scope: Fourier law, 3D heat conduction equation in Cartesian and cylindrical coordinates, Crit

Unit 2: Forced & Natural Convection

" Syllabus Scope: Boundary layer approximations, Nusselt, Reynolds, and Prandtl numbers, Empirical correl

ME304: Heat & Mass Transfer - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Thermal Radiation

" Stefan-Boltzmann law, Planck law, Wien displacement law, Emissivity, View factor algebra, Radiation exchange

Unit 4: Heat Exchangers & Mass Transfer

" Overall heat transfer coefficient, Fouling factor, LMTD method, Effectiveness-NTU method, Fick first law of

Unit 5: Boiling, Condensation & Mass Diffusion

" Pool boiling regimes, Critical heat flux, Dropwise and filmwise condensation, Mass transfer coefficients, Sh

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Thermal Conductivity Measurement of Insulating Material using Guarded Hot Plate Apparatus

Lab 2: Heat Transfer Coefficient Determination in Natural and Forced Convection through a Vertical Pipe

Lab 3: Stefan-Boltzmann Constant Verification using Radiation Test Apparatus

Lab 4: Performance Evaluation and Effectiveness Measurement of a Shell-and-Tube Heat Exchanger

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Frank P. Incropera, David P. DeWitt - Fundamentals of Heat and Mass Transfer Wiley 7th Ed

* J.P. Holman - Heat Transfer McGraw-Hill 10th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students